Advanced Topics in Statistics and Probability Theory

Applications to Cybersecurity and Computational Methods

Sapienza University of Rome
Master's Degree in Cybersecurity

Academic Year 2026/27

Abstract

This thesis explores fundamental concepts in statistics and probability theory, with a focus on theoretical foundations and practical applications in cybersecurity. We examine online algorithms for statistical computation, the Law of Large Numbers (LLN) and Central Limit Theorem (CLT) from both theoretical and practical perspectives, the formalization of probability through measure theory, methods for generating random variables, and statistical applications to cybersecurity problems. Each topic is presented with rigorous mathematical treatment alongside computational implementations.

Contents

1 Online Algorithms for Statistical Computation **2**
1.1 Introduction . . . 2
1.2 Welford's Online Algorithm for Mean and Variance . . . 2
1.2.1 Mathematical Foundation . . . 2
1.2.2 The Recurrence Relations . . . 2
1.2.3 Numerical Stability . . . 3
1.3 Online Quantile Estimation . . . 3
1.4 Applications to Cybersecurity . . . 3

2 Law of Large Numbers and Central Limit Theorem **3**
2.1 Probability Space Foundation . . . 3
2.2 Law of Large Numbers . . . 3
2.2.1 Weak Law of Large Numbers (WLLN) . . . 4
2.2.2 Strong Law of Large Numbers (SLLN) . . . 4
2.3 Central Limit Theorem . . . 4
2.3.1 Practical Implications . . . 4
2.4 Computational Verification . . . 5

3 Measure Theory and Probability Formalization **5**
3.1 Motivation . . . 5
3.2 Measure Theory Foundations . . . 5
3.3 Lebesgue-Stieltjes Integration . . . 6
3.3.1 Construction . . . 6
3.3.2 Lebesgue-Stieltjes Integration . . . 6
3.4 Applications to Probability . . . 6

4 Generation of Random Variables **6**
4.1 The Inverse Transform Method . . . 6
4.2 Examples . . . 7
4.2.1 Exponential Distribution . . . 7
4.2.2 Normal Distribution (Box-Muller) . . . 7
4.3 Rejection Sampling . . . 7

5 Applications to Cybersecurity **7**
5.1 Anomaly Detection via Statistical Thresholds . . . 7
5.2 Monte Carlo Simulation for Risk Assessment . . . 8
5.3 Randomness in Cryptography . . . 8
5.4 Statistical Independence Testing . . . 8

6 Conclusions **9**

1 Online Algorithms for Statistical Computation

1.1 Introduction

Online algorithms process data sequentially, updating statistics as new observations arrive without requiring storage of the entire dataset. These algorithms are crucial for streaming data analysis, big data applications, and real-time monitoring systems.

1.2 Welford's Online Algorithm for Mean and Variance

Traditional computation of sample variance requires two passes through the data: one to compute the mean, another to compute squared deviations. Welford's algorithm (1962) computes both mean and variance in a single pass using constant memory.

1.2.1 Mathematical Foundation

For a sequence of observations $x_1, x_2, \ldots, x_n$, the sample mean is:

$$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

The sample variance is:

$$s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2 \tag{2}$$

1.2.2 The Recurrence Relations

Welford discovered the following recurrence relations:

$$\bar{x}_n = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n} \tag{3}$$

$$M_n = M_{n-1} + (x_n - \bar{x}_{n-1})(x_n - \bar{x}_n) \tag{4}$$

$$s_n^2 = \frac{M_n}{n-1} \tag{5}$$

where $M_n = \sum_{i=1}^{n}(x_i - \bar{x}_n)^2$ is the sum of squared deviations.

Proof. Starting from the definition:

$$\begin{aligned}\bar{x}_n &= \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{n}\left[(n-1)\bar{x}_{n-1} + x_n\right] \\ &= \frac{n-1}{n}\bar{x}_{n-1} + \frac{x_n}{n} \\ &= \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n}\end{aligned}$$

For the variance update, we use the identity:

$$\begin{aligned}M_n &= \sum_{i=1}^{n}(x_i - \bar{x}_n)^2 \\ &= \sum_{i=1}^{n-1}(x_i - \bar{x}_n)^2 + (x_n - \bar{x}_n)^2\end{aligned}$$

Through algebraic manipulation (using the relation between $\bar{x}_n$ and $\bar{x}_{n-1}$), this reduces to the stated recurrence. □

1.2.3 Numerical Stability

Welford's algorithm is numerically stable because it avoids catastrophic cancellation that occurs in the two-pass formula $\sum x_i^2 - n\bar{x}^2$.

1.3 Online Quantile Estimation

Exact quantile computation requires sorting, which is not feasible for streaming data. P-square algorithm (Jain & Chlamtac, 1985) maintains estimates of quantiles using only five markers.

1.4 Applications to Cybersecurity

- **Network traffic monitoring**: Computing real-time statistics on packet sizes, connection durations
- **Anomaly detection**: Maintaining running mean and variance to detect outliers
- **Performance monitoring**: Tracking system metrics without storing complete histories

2 Law of Large Numbers and Central Limit Theorem

2.1 Probability Space Foundation

Definition 1 (Probability Space)**.** *A probability space is a triple* $(\Omega, \mathcal{F}, P)$ *where:*

- Ω *is the sample space (set of all possible outcomes)*
- $\mathcal{F}$ *is a* σ*-algebra on* Ω *(collection of events)*
- $P : \mathcal{F} \to [0, 1]$ *is a probability measure satisfying:*
  1. $P(\Omega) = 1$
  2. *For disjoint events* $A_1, A_2, \ldots$*:* $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

2.2 Law of Large Numbers

The Law of Large Numbers describes the convergence of sample averages to the expected value. Two versions exist: weak (convergence in probability) and strong (almost sure convergence).

2.2.1 Weak Law of Large Numbers (WLLN)

Theorem 1 (Weak Law of Large Numbers). *Let $X_1, X_2, \ldots$ be independent and identically distributed (i.i.d.) random variables with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then for any $\epsilon > 0$:*

$$\lim_{n\to\infty} P(|\bar{X}_n - \mu| > \epsilon) = 0 \tag{6}$$

Proof. We use Chebyshev's inequality: $P(|Y - E[Y]| \geq k) \leq \frac{\mathrm{Var}(Y)}{k^2}$.

Since $E[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$ (by independence):

$$\begin{aligned} P(|\bar{X}_n - \mu| \geq \epsilon) &\leq \frac{\mathrm{Var}(\bar{X}_n)}{\epsilon^2} \\ &= \frac{\sigma^2}{n\epsilon^2} \to 0 \text{ as } n \to \infty \end{aligned}$$

□

2.2.2 Strong Law of Large Numbers (SLLN)

Theorem 2 (Strong Law of Large Numbers). *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $E[|X_i|] < \infty$ and $E[X_i] = \mu$. Then:*

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1 \tag{7}$$

This is a stronger result: almost sure convergence implies convergence in probability, but not vice versa.

2.3 Central Limit Theorem

Theorem 3 (Central Limit Theorem). *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2 < \infty$. Define the standardized sum:*

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \tag{8}$$

Then Z_n converges in distribution to the standard normal distribution $N(0,1)$:

$$\lim_{n\to\infty} P(Z_n \leq z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2}\, dt \tag{9}$$

2.3.1 Practical Implications

The CLT explains why normal distributions appear ubiquitously:

- Measurement errors (sum of many small independent errors)
- Network latency (sum of processing times at multiple nodes)
- Physical phenomena involving many independent factors

Rule of thumb: For most distributions, $n \geq 30$ provides good normal approximation. For symmetric distributions, $n \geq 10$ often suffices.

2.4 Computational Verification

Consider generating n samples from an exponential distribution (highly skewed) and computing the sample mean. By CLT, the distribution of $\bar{X}_n$ should approach normal as n increases.

3 Measure Theory and Probability Formalization

3.1 Motivation

Classical probability (finite sample spaces with uniform probabilities) is insufficient for continuous distributions. Measure theory provides a rigorous foundation for:

- Defining probability on uncountable spaces (e.g., $\mathbb{R}$)
- Handling pathological sets and functions
- Unifying discrete and continuous probability
- Justifying operations like integration and differentiation of distributions

3.2 Measure Theory Foundations

Definition 2 (σ-Algebra)**.** *A collection $\mathcal{F}$ of subsets of Ω is a σ-algebra if:*

1. $\Omega \in \mathcal{F}$
2. *If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$*
3. *If $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$*

The σ-algebra defines which sets are measurable (events we can assign probabilities to).

Definition 3 (Measure)**.** *A measure μ on $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:*

1. $\mu(\emptyset) = 0$
2. *Countable additivity: For disjoint $A_1, A_2, \ldots \in \mathcal{F}$:*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

A probability measure is a measure with $\mu(\Omega) = 1$.

3.3 Lebesgue-Stieltjes Integration

3.3.1 Construction

The Lebesgue integral generalizes the Riemann integral by partitioning the *range* rather than the domain.

Definition 4 (Lebesgue Integral for Simple Functions)**.** *For a simple function* $s = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ *(where* $\mathbf{1}_{A_i}$ *is the indicator function):*

$$\int s \, d\mu = \sum_{i=1}^{n} a_i \mu(A_i) \tag{10}$$

For general measurable functions $f \geq 0$, approximate by simple functions:

$$\int f \, d\mu = \sup \left\{ \int s \, d\mu : s \text{ simple}, 0 \leq s \leq f \right\} \tag{11}$$

3.3.2 Lebesgue-Stieltjes Integration

Given a cumulative distribution function $F : \mathbb{R} \to [0, 1]$, the Lebesgue-Stieltjes integral with respect to F is:

$$\int f(x) \, dF(x) \tag{12}$$

This unifies discrete and continuous distributions:

- If F is absolutely continuous with density f: $\int g \, dF = \int g(x) f(x) \, dx$
- If F has jump discontinuities: $\int g \, dF = \sum_i g(x_i) p_i$ (discrete case)

3.4 Applications to Probability

Expected value: For random variable X with distribution F:

$$E[X] = \int x \, dF(x) \tag{13}$$

This formula works for discrete, continuous, and mixed distributions.

4 Generation of Random Variables

4.1 The Inverse Transform Method

The fundamental problem: Given a source of uniform random numbers $U \sim \text{Uniform}(0, 1)$, generate samples from an arbitrary distribution F.

Theorem 4 (Inverse Transform)**.** *Let* F *be a cumulative distribution function and* F^{-1} *its quantile function (generalized inverse). If* $U \sim \text{Uniform}(0, 1)$*, then:*

$$X = F^{-1}(U) \tag{14}$$

has CDF F*.*

Proof. For any x:

$$\begin{aligned} P(X \leq x) &= P(F^{-1}(U) \leq x) \\ &= P(U \leq F(x)) \quad \text{(by monotonicity of } F\text{)} \\ &= F(x) \quad \text{(since } U \sim \text{Uniform}(0,1)\text{)} \end{aligned}$$

□

4.2 Examples

4.2.1 Exponential Distribution

For $X \sim \text{Exp}(\lambda)$ with CDF $F(x) = 1 - e^{-\lambda x}$:

$$\begin{aligned} U &= 1 - e^{-\lambda X} \\ X &= -\frac{1}{\lambda}\ln(1-U) = -\frac{1}{\lambda}\ln(U) \end{aligned}$$

4.2.2 Normal Distribution (Box-Muller)

The inverse transform is not practical for normal distributions (no closed form for Φ^{-1}). Instead, use Box-Muller transformation:

Given $U_1, U_2 \sim \text{Uniform}(0,1)$ independent:

$$Z_1 = \sqrt{-2\ln U_1}\cos(2\pi U_2) \tag{15}$$

$$Z_2 = \sqrt{-2\ln U_1}\sin(2\pi U_2) \tag{16}$$

Then $Z_1, Z_2 \sim N(0,1)$ independently.

4.3 Rejection Sampling

When inverse transform is infeasible, rejection sampling generates from f using a proposal distribution g:

Algorithm 1 Rejection Sampling

1: Choose M such that $f(x) \leq Mg(x)$ for all x
2: **repeat**
3: Generate $Y \sim g$
4: Generate $U \sim \text{Uniform}(0,1)$
5: **until** $U \leq \frac{f(Y)}{Mg(Y)}$
6: **return** Y

Acceptance probability is $1/M$, so efficiency requires M close to 1.

5 Applications to Cybersecurity

5.1 Anomaly Detection via Statistical Thresholds

Using CLT and normal approximation for detecting unusual network behavior:

1. Collect baseline data: packet rates, connection durations, payload sizes
2. Compute mean μ and standard deviation σ using Welford's algorithm
3. For new observation x, compute z-score: $z = \frac{x-\mu}{\sigma}$
4. Flag as anomaly if $|z| > 3$ (approximately $p < 0.003$ under normality)

5.2 Monte Carlo Simulation for Risk Assessment

Estimate probability of rare events (e.g., system compromise) through simulation:

1. Model attack paths as random variables with known distributions
2. Generate random scenarios using inverse transform/rejection sampling
3. Estimate success probability as proportion of successful attacks
4. Use LLN to justify convergence of estimate

5.3 Randomness in Cryptography

Cryptographic security depends on high-quality random number generation:

- **Key generation**: Uniform sampling from key space
- **Initialization vectors**: Random seeds for encryption algorithms
- **Salting**: Random values to prevent rainbow table attacks

Statistical tests (NIST SP 800-22) verify randomness quality by checking deviation from theoretical distributions predicted by CLT and other probabilistic properties.

5.4 Statistical Independence Testing

In intrusion detection, testing whether user behavior (X) is independent of time-of-day (Y):

- Construct contingency table of (user actions) $\times$ (time periods)
- Compute expected frequencies under independence
- Use χ^2 test to detect dependence
- Dependence may indicate compromised accounts (unusual access patterns)

6 Conclusions

This thesis has examined fundamental concepts bridging theoretical probability and practical computation:

1. **Online algorithms** enable efficient computation of statistics in streaming environments, essential for real-time monitoring.
2. **LLN and CLT** provide theoretical justification for statistical inference and explain the ubiquity of normal distributions.
3. **Measure theory** rigorously formalizes probability, enabling treatment of continuous distributions and pathological cases.
4. **Random variable generation** transforms uniform randomness into arbitrary distributions, powering simulation and cryptography.
5. **Cybersecurity applications** demonstrate practical value of statistical theory in anomaly detection, risk assessment, and security testing.

The interplay between theory and computation is central to modern statistics: rigorous mathematical foundations ensure correctness, while efficient algorithms enable practical application at scale. In cybersecurity, where adversaries exploit both theoretical weaknesses and implementation flaws, this dual perspective is essential for building robust defenses.

References

[1] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419-420.

[2] Billingsley, P. (2012). *Probability and Measure* (Anniversary Edition). Wiley.

[3] Durrett, R. (2019). *Probability: Theory and Examples* (5th ed.). Cambridge University Press.

[4] Devroye, L. (1986). *Non-Uniform Random Variate Generation*. Springer.

[5] Feller, W. (1968). *An Introduction to Probability Theory and Its Applications*, Vol. 1 (3rd ed.). Wiley.

[6] Ross, S. M. (2014). *Introduction to Probability Models* (11th ed.). Academic Press.

[7] Jain, R., & Chlamtac, I. (1985). The P-square algorithm for dynamic calculation of quantiles and histograms without storing observations. *Communications of the ACM*, 28(10), 1076-1085.

[8] NIST (2010). *A Statistical Test Suite for Random and Pseudorandom Number Generators for Cryptographic Applications*. Special Publication 800-22 Rev. 1a.